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Synthetic Variable-Baseline Variable-Orientation Stereo Camera System for Relative Navigation in Close Proximity

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Abstract

Given the high number of inactive satellite orbiting and polluting the space around the Earth, In-Orbit servicing missions will play a substantial role in the industry in the upcoming decades. These missions could potentially revive numerous satellites through refueling or minor repairs, thereby reactivating valuable assets in space. Among the various techniques envisioned as possible solutions to the problem, the use of Space Manipulator Systems (SMS), i.e. servicing spacecrafts mounting manipulators on-board, is particularly interesting, as it offers great flexibility. Designing this kind of missions, however, is often challenging. As the system moves in close proximity of the target spacecraft, correctly estimating its position and attitude is paramount in order to avoid undesirable collisions. On this matter, studies have shown that relying on the direct kinematics of the manipulator to reconstruct the pose of the end-effector as a function of the spacecraft base pose and joint variables of the arm lacks robustness and does not meet the error thresholds required to enable this kind of technology. This problem can be mitigated using the direct measurements of a sensor mounted in the vicinity of the end-effector of the manipulator. While lidar sensors and stereo cameras necessitate higher power and mass budgets, opting for monocular cameras can mitigate this issue, albeit at the cost of reduced accuracy. This paper proposes a novel approach in which an UnScented QUaternion Estimator (USQUE) is used to concurrently perform non-linear filtering on the attitude and position of both the base of the chasing spacecraft and the end-effector of the manipulator within a fully relative framework. This is achieved exclusively through information derived from two cameras mounted, respectively, on the spacecraft base and the manipulator end-effector, effectively creating a variable-baseline stereo camera system. The results demonstrate the efficacy of the method, showcasing high accuracy in the reconstruction of the position and orientation of both the reference frames of interest. As a secondary outcome, a high-fidelity model of the Space Manipulator System (SMS) is generated using Modelica. This model also incorporates a fully relative representation of gravitational acceleration in the Local-Vertical Local-Horizontal (LVLH) frame centered on the target. This approach circumvents potential numerical challenges stemming from the significantly smaller distance between the target and servicing spacecraft compared to the one from the center of gravitational attraction. Additionally, it preserves the possibility to incorporate gravitational perturbations into the modeling, a consideration often overlooked in the related literature.

Keywords: Computer Vision, In-orbit servicing, Relative Navigation, Stereo Vision.

1. Introduction

One of the most pressing challenges in designing new space missions is represented by the overcrowding of orbits due to the increasing number of Resident Space Objects (RSOs). This creates higher risks for both new and existing satellites and complicates access to space. While several mitigation guidelines have been proposed [1, 2], there remains a critical need for In-Orbit Servicing (IOS) to refuel, repair, and remove compromised assets from orbit.

Robotic manipulators mounted on servicing spacecraft, known as Space Manipulator Systems (SMS), play a crucial role in this endeavour. Early systems, such as the Shuttle Remote Manipulator System (Canadarm), demonstrated the effectiveness of robotic arms for satellite operations [3]. More advanced systems like the European Robotic Arm (ERA) and Dextre are now essential for maintaining the International Space Station (ISS) [4].

A key challenge in these missions is navigating and controlling the robotic manipulator during the final approach to the target. Due to communication delays and limited control windows, autonomous navigation and control algorithms are a necessity to enhance mission success [5, 6, 7].

Moreover, precise positioning is critical and is usually achieved with sensors like LIDAR or radar payloads.

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However, these technologies have certain limitations, such as high power and mass budget consumption, and sensitivity to interference [8, 9]. Another viable alternative is represented by stereo vision systems, however they require complex image processing and are affected by lighting conditions [10, 11, 12]. In addition, relying solely on one type of sensor also introduces single-point-of-failure in the design [13].

To overcome these issues, multi-sensor approaches are necessary for more reliable and accurate navigation. Monocular cameras, which are lightweight and energy-efficient, provide an alternative solution. Several navigation techniques using monocular cameras have already been implemented. Feature-based methods like SIFT and ORB are common, though they each have strengths and weaknesses [14]. Optical Flow and Structure from Motion (SfM) are also useful but can be computationally demanding [15, 16]. Pose estimation techniques, such as PnP, can be highly accurate but depend on a reliable 3D model of the target [17, 18].

Despite these advantages, monocular cameras do not provide direct depth information, complicating distance measurements. This often requires additional sensors or more sophisticated algorithms, increasing system complexity [19]. They also struggle with varying lighting and occlusion [20], and their computational demands can be challenging for real-time space applications [21, 22].

Additionally, much of the research on IOS missions and SMSs underrepresents environmental disturbances and simplifies the interaction between the spacecraft's motion and the manipulator's movements [23, 24, 25], limiting its applicability in more dynamic mission scenarios.

This paper proposes using a variable geometry camera system, with two cameras mounted on both the spacecraft and the manipulator, to offer a more robust navigation approach. This setup provides continuous monitoring, reduces the likelihood of occlusion, and improves adaptability in dynamic environments [26]. By combining advanced image processing techniques like feature tracking and machine learning, this configuration enhances navigation accuracy [27].

The following sections introduce the different methodology adopted in this work for modeling the SMS and the simulation architecture used. The adopted filtering technique will then be briefly discussed. Subsequently, detailed analyses will be discussed and possible future developments will then be presented.

2. Space Manipulator System modeling

First, a simulator for the SMS is implemented (Fig. 1). This simulator uses a simplified open-loop control algorithm to track a prescribed relative trajectory with both the base and the end effector of the SMS. The control leverages the SMS inverse dynamics to infer the control actions to be exerted on the base of the spacecraft and on the joints of the manipulator in order to follow the prescribed guidance. In particular, the relative trajectory is designed to represent typical IOS relative trajectory [28].

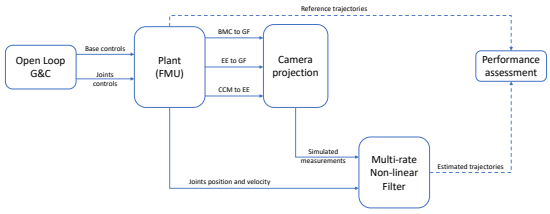


Fig. 1: Overall simulation and testing architecture.

These determined controls are then fed to a Functional Mockup Unit (FMU) describing a high-fidelity model of the multibody system representing the SMS. The FMU then provides the evolution in time of the pose of the end-effector (and of the cameras) with respect to the body frame of the target RSO which can be used as a ground truth for the navigation filter. Moreover, the poses output by the FMU can be used to generate synthetic camera measurements by leveraging the knowledge of the target model. These measurements, together with the manipulator joints' positions and velocities, are then leveraged in the correction step of the navigation filter. The information from the joints is considered to be known without uncertainty in the context of this study.

While the FMU represents the accurate model of the dynamics, a simplified description of the SMS is needed to use the navigation filter. The two models are described in the next subsections.

2.1 Simulator Dynamics

As already mentioned, the high fidelity model is constituted by an FMU which is created using Modelica[1] (and the Dymola Modelica tool[2]). This tool supports the FMI standard, allowing the execution of FMU or the creation of new ones that can be executed from within other tools supporting FMI (here Simulink is used to execute the FMU). Fig. 2 shows the full model for the simulation.

The Modelica library allows to model the high-level SMS by combining subsystems of the controlling assembly of the joints including sensors and motors. The details of the implementation of each element of the manipulator mechanical assembly is considered outside the scope of this paper.

Accelerations on the standard Body objects from the Modelica library can only be expressed in the "world" model (i.e., geocentric inertial frame). To overcome this

[1] https://modelica.org/ [last accessed on July, 17th, 2024]
[2] https://www.3ds.com/products/catia/dymola [last accessed on July, 17th, 2024]

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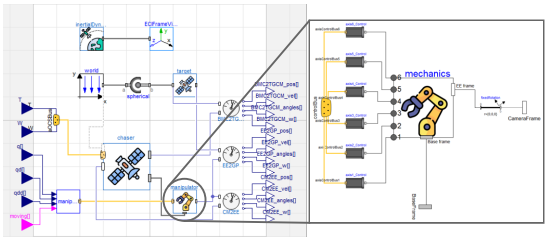


Fig. 2: Full Modelica model of the chasing spacecraft system and dynamical environment (left) and detail of the manipulator model (right).

shortcoming, we built a custom subclass of the standard body object and implemented a dynamic acceleration block capable of providing accelerations for relative motion between satellites [29].

2.2 Filter Dynamics

A simplified SMS model is adopted for the navigation filter predictions. By using the Lagrangian approach, such dynamics can be reformulated as[30, 31]:

$$\begin{bmatrix} M_0 & M_{0m} \\ M_{m0} & M_m \end{bmatrix} \begin{bmatrix} \ddot{\theta}_0 \\ \ddot{\theta}_m \end{bmatrix} + \begin{bmatrix} C_b & C_{0m} \\ C_{m0} & C_m \end{bmatrix} \begin{bmatrix} \dot{\theta}_0 \\ \dot{\theta}_m \end{bmatrix} + \begin{bmatrix} N_0 \\ N_m \end{bmatrix} - \begin{bmatrix} \tau_0 \\ \tau_m \end{bmatrix} = \tag{1}$$

$$= M\ddot{\theta} + C\dot{\theta} + N - \tau = 0 \tag{2}$$

Where θ is the vector of generalized coordinates associated to the base (θ_0) and manipulator (θ_m) respectively. Consequently, the matrix M is used to represent the Generalized Inertia Matrix (GIM), C is the Convective Inertia Matrix (CIM) and N accounts for the conservative perturbations (e.g., gravitational potential).

A typical SMS differs from standard ground-based manipulators in the fact that the base is not constrained. This causes the off-diagonal terms of the GIM and the CIM to be non-zero. Therefore, forces applied on the manipulator (including disturbances) are transmitted to the base.
Moreover, Eq. (1) is too complex ($O(n^3)$ with n links) to be used for simulation purposes. In this work, we adopt the Screw Theory and the Articulated Body Model (ABM) [30] for the representation of the manipulator through the SPART software [32]. This is a recursive technique of lower complexity ($O(n)$).
With SPART, the accelerations of the generalized coordinates related to the base and end-effector, $\ddot{\theta}_0$ and $\ddot{\theta}_m$, can be expressed as a function of the control actions applied, τ_0 and τ_m, the forces (wrenches) acting on the system, W_{F_0} and W_{F_m}, and the current configuration and velocities, θ_0, θ_m, $\dot{\theta}_0$ and $\dot{\theta}_m$.
The state-space representation of the system can therefore

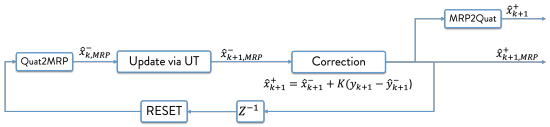


Fig. 3: Schematic representation of an UnScented QUaternion Estimator. Here, x represents the states, y the measurements, k the time instant, and K the Kalman gain. The minus and plus signs refer to quantities before and after the correction step, respectively.

be reformulated as:

$$\begin{cases} \dot{q}_0 = \frac{1}{2}[w_0; 0] \otimes q_0 \\ \dot{r}_0 = v_0 \\ [\dot{w}_0, \dot{v}_0] = \ddot{\theta}_0 \\ \dot{q}_{EE} = \frac{1}{2}[w_{EE}; 0] \otimes q_{EE} \\ \dot{r}_{EE} = v_{EE} \\ [\dot{w}_{EE}, \dot{v}_{EE}] = J_{0_{EE}}\ddot{\theta}_0 + \dot{J}_{0_{EE}}\dot{\theta}_0 + J_{m_{EE}}\ddot{\theta}_m + \dot{J}_{m_{EE}}\dot{\theta}_m \end{cases} \tag{3}$$

with quaternions q_0 and q_{EE} representing the orientation of the base and of the end-effector. Similarly, r_0, r_{EE}, v_0 and v_{EE} represent the position and velocity of the two components. Differently from what has already largely been done in the literature, the state definition is here based on the end-effector quantities rather than on the joints ones. This causes the flexibility of the joints, and therefore the uncertainty on their displacement, to be implicitly accounted for in the filter and hence increase its robustness and improve its performance.

3. Navigation Filter

In this work, we leveraged the Unscented Kalman Filter (UKF) for navigation in order to avoid common pitfalls of its linearized counterpart. Indeed, it is essential in such a complex simulation to retain the non-linearity during the filter propagation step [33]. Due to the adoption of quaternions to describe the attitude we had to resort to its multiplicative variant, the UnScented QUaternion Estimator (USQUE), to preserve rotation properties during update and correction steps of the filter [34]. To achieve this, this filter keeps the orientation error, represented through Generalized Rodriguez Parameters [35], as an internal state of the filter rather that using the estimated attitude quaternion. A simple schematic representation of this architecture is reported in Fig. 3.

Concerning the measurements that are provided to the filter, the approach followed in this paper uses target features extracted from images as measurements and it consists of two main steps. After extracting features from images, the raw pixel measurements are embedded directly

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in the filter by adopting a simple monocular camera model for prediction of features location. The correction step of the filter directly compares these to the actual measurements coming from the sensors to correct the state. This approach is in contrast with state-of-the-art methods for pose estimation that leverage pixel information to reconstruct the pose and treat this information as measurements in the filter [21, 36, 37, 11]. Therefore the currently implemented approach may be further improved by adopting these methods.

4. Results

4.1 Case Study Description

In this study, a simplified target RSO is used. This is represented by a rectangular prism whose corners are taken as reference features to be leveraged by the navigation filter. Also, reasonable illumination conditions that may be required for operations with a passive optical payload chasing an RSO in a typical LEO Sun-Synchronous Orbit are adopted. As a consequence, the number of visible markers can vary in time. Moreover, the base of the chaser is assumed to be moving with non-uniform velocity in a straight line at distances from the target that range from 12 to 20 meters, coherently with the last stage of a IOS mission. Conversely, the guidance that the FMU of the manipulator needs to track has been determined by using a minimum-jerk trajectory between some predetermined keypoint poses. The RSO attitude is determined by assuming an initial spin rate around one of its axes (in the range of 0.5 to 1 deg/s). To conclude, the measurement rate is assumed to be 1Hz and with a 1-pixel standard deviation in order to account for effects of non ideal image processing.

4.2 Results

The results for the simulation are reported in Fig. 4. This approach demonstrates the efficacy of the method in reconstructing both base and end-effector poses. The pointing error is also particularly low, indicating an efficient reconstruction of the relative orientation of the base and end-effector despite the large initialization errors. This highlights how the approach effectively mitigates the lack of a direct range measurement and how the two-camera system, coupled with a suited navigation algorithm, effectively constitutes a variable-baseline variable-orientation stereo camera.

To verify the consistency of the filter, the innovation of the filter during execution has been reported in Fig. 5. This quantity appears to be randomly distributed around zero and much smaller in magnitude with respect to the typical dimension of the measurements (i.e., 1200-1920 pixels); the cross-correlation of the innovation is also zero-mean and small in magnitude, as expected.

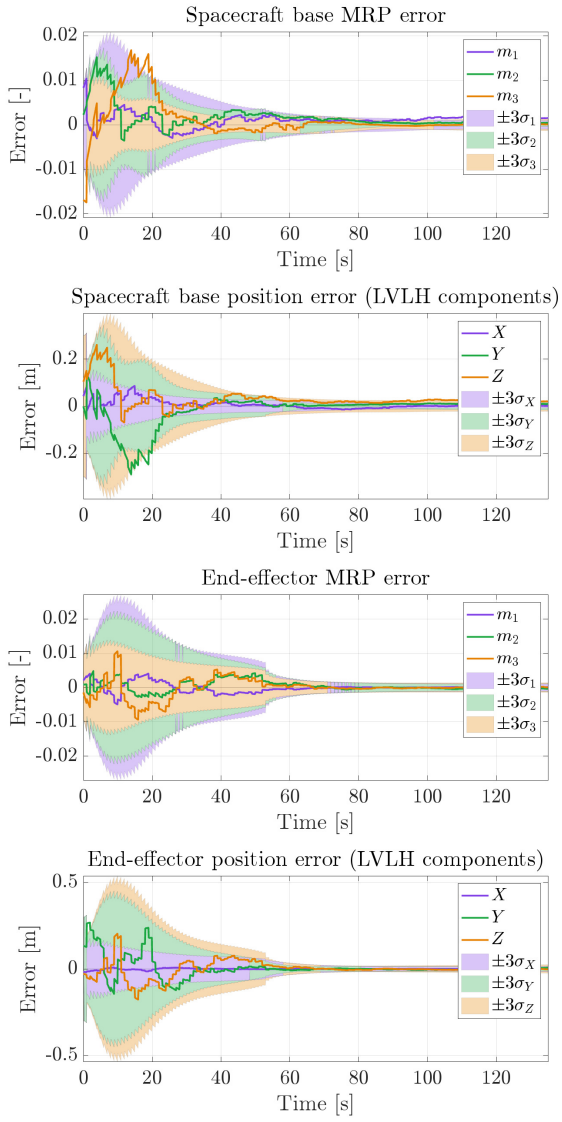


Fig. 4: Pose reconstruction errors of the filter.

5. Conclusions

The presented paper demonstrated that by adding two monocular cameras on a SMS and allowing one to move with the manipulator's end-effector it is possible to obtain a variable-baseline and variable-orientation stereo camera system that removes the need for a depth estimation for navigating in proximity of a target RSO. This approach offers enhanced reliability, accuracy, and flexibility as it is testified by the reduced errors in the pose estimation of the base and end-effector, which is in the order of millimeters and tenths of a degree in the envisioned test case. This paper adopted a simulator FMU to retrieve the measurements for the navigation algorithm. Future

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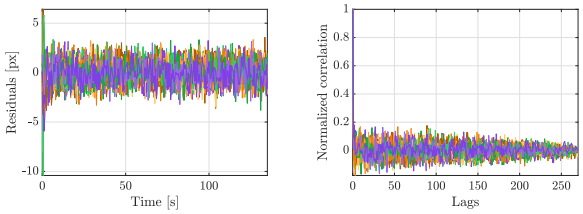


Fig. 5: Innovation (left) and normalized cross-correlation of the innovation (right). Each line refers to one of the measurement components.

development include a more refined simulation or real data collected in the laboratory. Moreover, in presence of a more complex target with larger number of features, future works may also investigate criteria for the selection of the best subset of measurements, to keep the dimensionality of the problem within reasonable limits while ensuring minimal loss of accuracy.

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